

Coding & Solution

Date	10 July 2026
Team ID	xxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python,HTML5,CSS3,JavaScript
Framework(s) Used	Flask,Scikitlearn,XGBoost,Pandas,Numpy,Matplotlib,Seaborn,Bootstrap,Chart.js,Joblib

Key Features Implemented	☐ Historical flood prediction using Machine Learning ☐ Random Forest classification model ☐ Data preprocessing and feature scaling ☐ Interactive Flask web application ☐ Weather parameter input validation ☐ Flood risk prediction with confidence score ☐ Feature contribution visualization ☐ Responsive Glassmorphism UI ☐ Model serialization using Joblib ☐ IBM Cloud deployment ready
Pending / Incomplete Features	None
Setup / Run Instructions	1. Install Python 3.10 or above. 2. Install dependencies using: pip install -r requirements.txt 3. Run the application: python app.py 4. Open: http://localhost:8080 5. Enter meteorological readings and click Predict to view flood risk.



Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes

6	Code is committed to a version control repository	Yes
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Additional Notes / Comments:

The project follows a modular architecture with separate folders for machine learning models, utility functions, frontend templates, static assets, and application logic. A Random Forest classifier is used for flood prediction, while Flask provides the backend services. The application generates confidence scores and feature-based explanations for each prediction, making the solution reliable, maintainable, and suitable for deployment on IBM Cloud.